

Assignment 2 – Group 5

Financial Engineering: Risk Management

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Introduction

This report analyzes a corporate bond portfolio managed by Polimi Capital on February 15, 2008, consisting of two fixed-rate bonds issued by Beta (€ 10M face value each). We derive Z-spreads and default intensities, study a credit deterioration scenario, and compare market-implied default probabilities with historical estimates from a rating transition matrix. Finally, we compute the VaR of a 100-bond portfolio via Monte Carlo simulation with a single-factor model. The risk-free curve is bootstrapped from Assignment 0.

To execute these tasks, we utilized and corrected two main Python modules:

- `date_functions.py`: handles day-count conventions and payment schedules.
- `ex2_utilities.py`: This module contains the core functions for pricing defaultable bonds. It implements cash flow generation and includes specific pricing formulas: `dirty_price_from_zspread` to calculate prices using a constant Z-spread, and `defaultable_bond_dirty_price_from_intensity` to compute prices based on both constant and piecewise-constant hazard rates.

Case Study A

Task 1: Constant Z-Spread

We assume a constant Z-spread z over the life of each bond. The modified discount factor $\hat{B}(t, T)$ and the theoretical dirty price $\bar{C}(t)$ are defined as:

$$\hat{B}(t, T) = B(t, T) \exp \left(- \int_t^T z(s) ds \right), \quad \bar{C}(t) = c \sum_{n=1}^N \delta_{n-1} \hat{B}(t, t_n) + \hat{B}(t, t_N)$$

Since z is constant, the integral simplifies to $z(T - t)$. For both the 1-year and 2-year bonds, we use `fsolve` to numerically find the spread z that equates the theoretical price $\bar{C}(t)$ to the observed market price.

Bond	Z-Spread
1-year	1.4722%
2-year	1.5187%

The 2-year bond yields a slightly higher Z-spread, reflecting the increased credit risk associated with a longer maturity.

Task 2: Constant Default Intensity

We assume a constant default intensity λ and a recovery rate $\pi = 40\%$. The survival-adjusted discount factor $\bar{B}(t, T)$ and the theoretical dirty price $\bar{C}(t)$ are defined as:

$$\bar{B}(t, T) = B(t, T) \exp \left(- \int_t^T \lambda(s) ds \right), \quad \bar{C}(t) = \bar{c} \sum_{n=1}^N \delta_{n-1} \bar{B}(t, t_n) + \bar{B}(t, t_N) + \pi \sum_{n=1}^N e(t; t_{n-1}, t_n)$$

Since λ is constant, the integral simplifies to $\lambda(T - t)$. For each bond, we use `fsolve` to find the independent intensity λ that equates the theoretical price $\bar{C}(t)$ to the observed market dirty price.

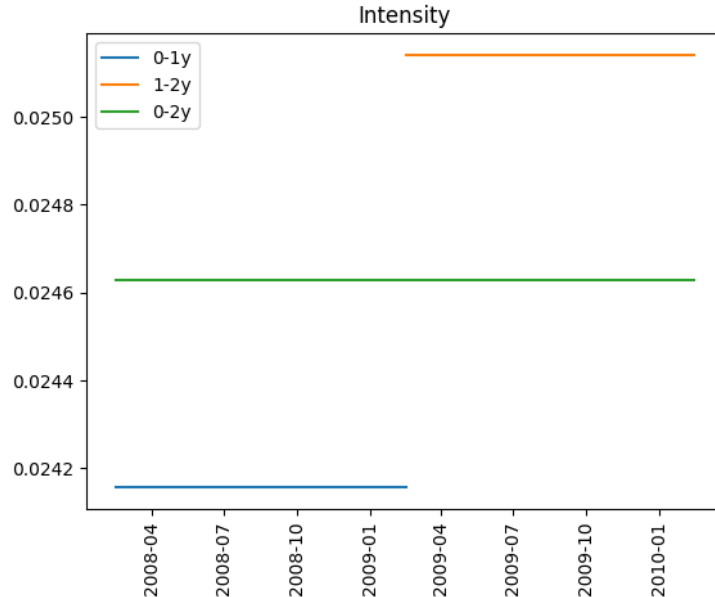
Bond	Intensity λ	Survival Prob.	Default Prob.
1-year	2.4157%	97.6262%	2.3738%
2-year	2.4628%	95.2130%	4.7870%

The two intensities are not equal. Since we price each bond independently, a flat λ calibrated to a 1-year maturity naturally differs from the 2-year one. This difference reflects the term structure of credit risk. However, their very close values (2.41% and 2.46%) indicate that the market perceives the issuer's credit risk as relatively stable over the two-year horizon, resulting in an almost flat term structure of default risk.

Task 3: Piecewise-Constant Default Intensity

We apply the same pricing formula as in Task 2, but assume the default intensity λ is piecewise-constant. We bootstrap the curve sequentially: λ_1 is calibrated from the 1-year bond, and λ_2 (valid strictly from year 1 to year 2) is extracted from the 2-year bond by keeping λ_1 fixed.

Period	Intensity	Horizon	Default Prob.
[0, 1y]	2.4157%	1-year	2.3738%
[1y, 2y]	2.5141%	2-year	4.7911%



The piecewise-constant approach gives a better approximation of Beta's creditworthiness. While a single constant intensity averages the risk over two years, the piecewise method isolates the exact marginal default risk of the second year. This prevents the short-term expectations (year 1) from contaminating the longer-term ones (year 2).

Task 4: Creditworthiness Deterioration scenario (Scenario 1)

We analyze a credit deterioration scenario where the 2-year bond's dirty price drops to € 97, while the 1-year bond remains at € 100. We re-bootstrap the piecewise-constant intensities using the same sequential approach as in Task 3.

Marginal Intensity (λ)			Default Prob.		
Period	Base	Scenario 1	Horizon	Base	Scenario 1
[0, 1y]	2.4157%	2.4157%	1 Year	2.3738%	2.3738%
[1y, 2y]	2.5141%	13.6003%	2 Years	4.7911%	14.7562%

The results show a major change in market expectations. Since the 1-year bond price did not move, the risk for the first year stays exactly the same (2.37%). However, to justify the lower price of the 2-year bond, the model must increase the risk for the second year.

In this scenario, the marginal intensity for the second period jumps to 13.56%. This tells us that the market is not worried about an immediate collapse, but expects a very high risk of default between the first and the second year. By using a piecewise-constant model, we can isolate this "future shock" without affecting the short-term credit profile.

Task 5: Comparison with real world probability

Using the rating transition matrix at one year (with $\pi_{IG \rightarrow Def} = 2\%$), we compute historical (real-world) default probabilities. The 2-year transition matrix is obtained as $M^{(2)} = M \cdot M$. We compare with the market-implied estimates under Scenario 1 (Task 4). The conditional default probability is computed as:

$$P(\tau \in [1y, 2y] \mid \tau > 1y) = \frac{P_D^{2y} - P_D^{1y}}{1 - P_D^{1y}}$$

Measure	Real-World	Implied (Scenario 1)
$P(\text{default by 1y})$	2.00%	2.37%
$P(\text{default by 2y})$	4.71%	14.76%
$P(\text{def in } [1y, 2y] \mid \text{surv. to 1y})$	2.77%	12.68%

The implied conditional probability under Scenario 1 (12.68%) is about five times the historical one (2.77%). This gap has two drivers: first, bond prices include a credit risk premium that pushes risk-neutral probabilities above real-world ones; second, the price drop to €97 signals that the market expects Beta's credit quality to worsen significantly in the second year, something the historical transition matrix cannot reflect.

Case Study B: Single-Factor Model simulations

Task 6a: Portfolio Mark-to-Market

The portfolio consists of 100 zero-coupon bonds with 2-year maturity, each paying €1M at maturity, issued by IG-rated corporations. We price each bond using real-world default probabilities from the transition matrix. Default can occur in two periods, and recovery π is received at the midpoint of the period in which default occurs:

$$\bar{B}^{IG}(t_0, t_2) = \underbrace{(1 - P_D^{2y}) B(t_0, t_2)}_{\text{survival leg}} + \pi \left[\underbrace{P_D^{1y} B(t_0, t_{0.5})}_{\text{default in } [0, 1y]} + \underbrace{(P_D^{2y} - P_D^{1y}) B(t_0, t_{1.5})}_{\text{default in } [1y, 2y]} \right]$$

Description	Value
Portfolio MtM	€90,175,660.67

Task 6b: 1-Year Forward Prices of the ZCB

Building on the framework from Task 6a, we evaluate the 1-year forward price of the ZCB, conditional on the issuer's rating at $t_1 = 1$.

The conditional forward price $\bar{B}(t_1, t_2)$ maintains the same structure as the spot price. Denoting the forward risk-free discount factors as $B(t_1, T) = \frac{B(t_0, T)}{B(t_0, t_1)}$, the pricing formula becomes:

$$\bar{B}^{Rating}(t_1, t_2) = \underbrace{(1 - P_{Rating \rightarrow Def}) B(t_1, t_2)}_{\text{survival leg}} + \underbrace{\pi P_{Rating \rightarrow Def} B(t_1, t_{1.5})}_{\text{default in } [1y, 2y]}$$

Using the conditional 1-year default probabilities from the transition matrix ($P_{IG \rightarrow Def} = 2\%$ and $P_{HY \rightarrow Def} = 5\%$), we obtain the forward prices summarized in Table.

Rating Scenario at t_1	Default Prob. (P_D)	Forward ZCB Price $\bar{B}(t_1, t_2)$
Investment Grade (IG)	2%	€953405.20
High Yield (HY)	5%	€936256.83

Table 1: Conditional 1-Year Forward Prices

Task 6c: Monte Carlo Simulation

We simulate the single-factor model $v_i = \rho y + \sqrt{1 - \rho^2} \epsilon_i$, where $y \sim \mathcal{N}(0, 1)$ is the systematic factor common to all issuers and $\epsilon_i \sim \mathcal{N}(0, 1)$ are i.i.d. idiosyncratic shocks, with 1,000,000 scenarios and three correlation levels ($\rho = 0\%, 25\%, 50\%$). Each issuer's state is determined by comparing v_i against thresholds $\Phi^{-1}(P_{Def})$ and $\Phi^{-1}(P_{Def} + P_{HY})$ derived from the transition matrix. We evaluate the VaR under two loss definitions: (i) defaults only, where only default events generate losses, and (ii) defaults and migrations, where downgrade and status-quo contributions are also included. The loss for each scenario is:

$$L = N_{def} \cdot \ell_{def} + N_{dg} \cdot \ell_{dg} + N_{sq} \cdot \ell_{sq}$$

where $\ell_{def} = \pi \cdot \frac{B(t_0, t_{0.5})}{B(t_0, t_1)} - \bar{B}(t_1, t_2)$ is the loss given default, $\ell_{dg} = \bar{B}^{HY}(t_1, t_2) - \bar{B}(t_1, t_2)$ is the loss given downgrade, and $\ell_{sq} = \bar{B}^{IG}(t_1, t_2) - \bar{B}(t_1, t_2)$ is the gain if the issuer remains IG.

Average Event Counts

The marginal probabilities are confirmed to be independent of correlation:

Correlation	Avg. Defaults	Avg. Downgrades	Avg. Status Quo
$\rho = 0\%$	2.0006	25.0048	72.9946
$\rho = 25\%$	2.0017	25.0068	72.9915
$\rho = 50\%$	2.0015	25.0092	72.9893

99% Value-at-Risk

Correlation	Expected Loss (EL)		VaR 99%	
	Def. Only	Def. + Mig.	Def. Only	Def. + Mig.
$\rho = 0\%$	€1,061,508.33	€427.84	€3,183,517.41	€2,166,007.95
$\rho = 25\%$	€1,062,078.18	€1,047.59	€4,244,689.88	€3,583,292.28
$\rho = 50\%$	€1,061,987.45	€995.42	€8,489,379.76	€8,258,159.83

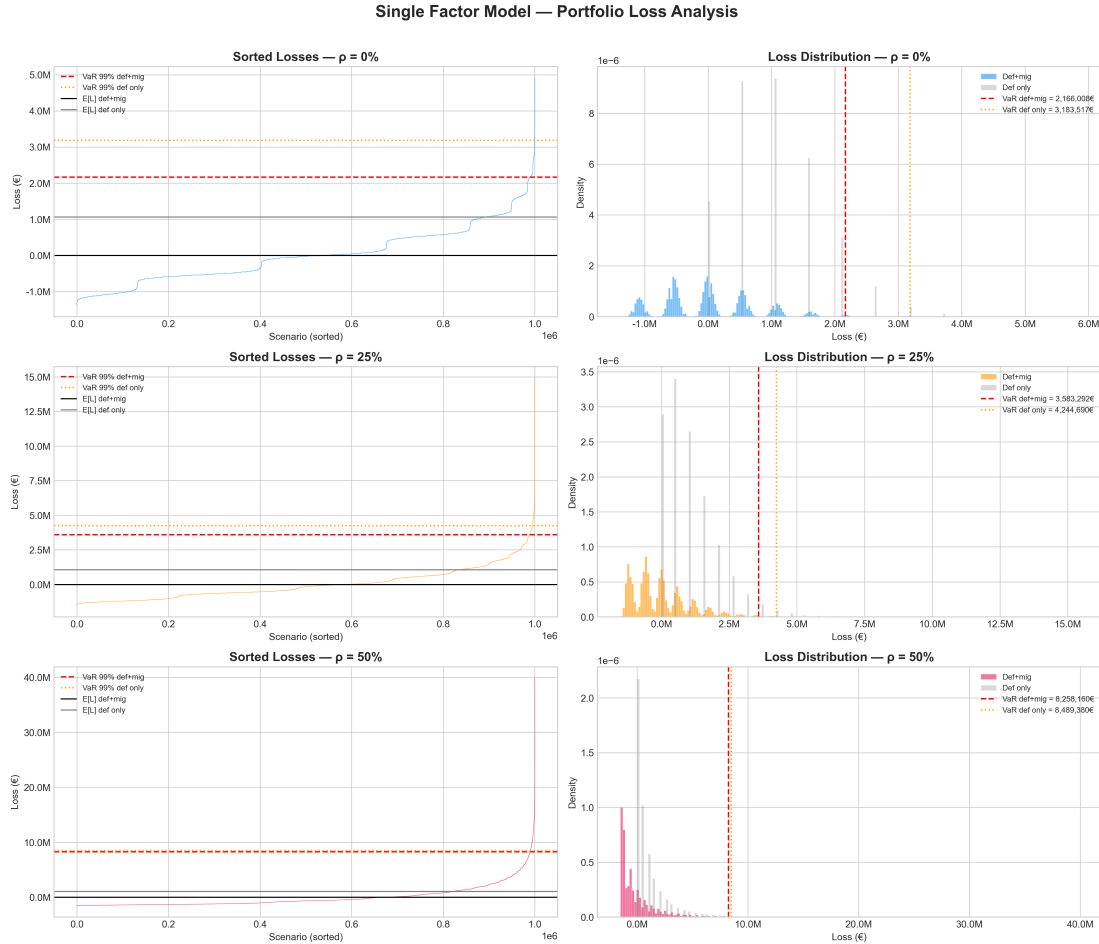


Figure 1: Portfolio loss distribution across correlation levels. Red dashed line: VaR 99% (defaults + migrations). Orange dotted line: VaR 99% (defaults only). Black line: expected loss.

Task 6d: VaR Ranking and Discussion

Effect of correlation. The expected loss is stable across correlations, but the 99% VaR increases sharply: from €2.2M ($\rho = 0\%$) to €8.3M ($\rho = 50\%$) in the full model. Higher correlation causes defaults and downgrades to cluster in the same scenarios, fattening the tail of the loss distribution.

Migration risk vs. default risk. The VaR under “defaults + migrations” is lower than the “defaults only” VaR. This is not because migration risk reduces losses, but because the two cases use different P&L constructions. In the full model, each scenario includes the gains from issuers that remain IG ($\ell_{sq} > 0$), which partially offset default and downgrade losses in the tail. In the defaults-only case, no such offset exists: the expected loss is ~€1M (vs. ~€0 in the full model), and the entire loss distribution is shifted to the right. The key takeaway is that migration risk is a meaningful additional source of tail risk, but its impact is partially masked by the diversification benefit of status-quo gains when all three states are modelled jointly.

Methodological Note on VaR Comparison

For a perfectly rigorous comparison between the two VaR scenarios, a strictly independent 2-state model (Default vs. Survival) should be implemented. First, we would define a 2×2 transition matrix where a bond either defaults (probability PD) or survives in its current state (probability $1 - PD$). Next, a new expected forward price must be calculated using only these two outcomes. The specific *Loss Given Default* and *Loss given Status Quo* would then be computed strictly relative to this new 2-state expected price. Finally, applying these adjusted P&L values in the Monte Carlo simulation would ensure an unbiased comparison against the 3-state VaR. However, we opted to maintain our original setup to keep the model simple and to focus primarily on the impact of rating migrations, accepting this minor approximation.

Identified Errors

According to the assignment instructions, we identified and corrected some errors in the notebooks. We also ensured that all libraries and functions were correctly imported from our own modules.

Errors in ex2_notebook

- **Coupon rate initialization:** Coupon rates must be expressed as percentages of the notional. The original initialization did not apply this correctly.
- **Incorrect function arguments (Q1 and Q2):** Both `defaultable_bond_dirty_price_from_z_spread` and `defaultable_bond_dirty_price_from_intensity` used `today` for both date arguments. The correct inputs are `settlement_date` and `today`.
- **2-year survival probability (Q3):** The expression

$$\text{surv_prob_2y} = \text{surv_prob_1y} \cdot e^{-h_{1y,2y} \cdot 1}$$

incorrectly used a 1-year interval. We have to consider the year fraction with ACT/365 day-convention (`yf_1y2y = year_frac_act_x(expiry1, expiry2, 365)`)

- **Transition matrix squaring (Q5):** The 2-year transition matrix must be computed via matrix multiplication, not elementwise one.
- **Forward price for IG issuer (Q6):** The formula used `survival_prob_hy_1y` instead of `survival_prob_ig_1y`.
- **Forward price with transitions (Q6):** The default leg is incorrectly multiplied by `fwd_df`. The correct term uses the ratio `df_1y_def / df_ref_date`.

- **Single-factor firm model (Q6):** The model was implemented as

$$v_i = \sqrt{\rho} y + \sqrt{1 - \rho} \varepsilon_i,$$

but the correct version is

$$v_i = \rho y + \sqrt{1 - \rho^2} \varepsilon_i.$$

- **Downgrade counting logic (Q6):** The notebook counted downgrades as `(v < barrier_downgrade)`, which incorrectly included defaults. The correct condition doesn't consider them.
- **VaR confidence level (Q6):** The code used `np.quantile(scenario, 0.001)`, corresponding to 99.9% VaR. For 99% VaR, the correct quantile is 0.01.
- **Discount factors:** We have to compute them with respect to the `settlement_date` and not today